VIKAS PURBIA

AI ENGINEER

Udaipur, India | +91 9166325693 | vikas.purbiaa@gmail.com | linkedin.com/in/vikas-purbia-10734432a | github.com/VIKAS-PURBIA

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building end-to-end machine learning systems: classical ML, deep learning (CNNs and transfer learning), NLP, and production-style Gen AI / LLM applications (RAG, LLM agents, vector search, evaluation harnesses). Strong in Python, SQL, TensorFlow, scikit-learn, FastAPI and Microsoft Azure, with measurable outcomes including 92% multiclass image classification accuracy, a 30% process-efficiency gain, and a fully evaluated local RAG support assistant with citations, refusal safety and human escalation. Seeking an AI Engineer role to design, evaluate, and ship reliable ML and LLM-powered products.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash

Machine Learning: scikit-learn, regression, K-Means clustering, Random Forest, RFM analysis, feature engineering, data preprocessing, model evaluation (accuracy, precision, recall, F1, confusion matrix), hyperparameter tuning

Deep Learning & Computer Vision: TensorFlow, Keras, Convolutional Neural Networks (CNN), transfer learning (MobileNetV2, EfficientNetB0), data augmentation

NLP & Generative AI: Natural Language Processing (NLP), sentiment analysis, Retrieval-Augmented Generation (RAG), embeddings, vector database (Qdrant), hybrid retrieval (BM25 + dense vectors, Reciprocal Rank Fusion), Large Language Models (LLMs), LLM agents and routing, prompt engineering, prompt-injection defense, Ollama (Llama 3.2, nomic-embed-text), LLM evaluation harnesses

Data Analysis & Visualization: pandas, NumPy, Matplotlib, Seaborn, Tableau, exploratory data analysis (EDA), statistical analysis

Cloud, MLOps & Tools: Microsoft Azure, FastAPI, REST APIs, Streamlit, Docker, Git, GitHub, pytest, CI workflows, ML pipelines, model deployment

WORK EXPERIENCE

AI/ML Intern — Labmentix Pvt Ltd, Udaipur, India

Jul 2025 – Sep 2025

- Engineered machine learning models that improved predictive accuracy for patient diagnosis and contributed to a reduction in processing errors.
- Partnered with cross-functional teams to redesign data processing strategies, achieving a 30% increase in efficiency.
- Leveraged Python and Tableau for advanced data analysis and visualization, enabling data-driven decision-making among stakeholders.

PROJECTS

Support Copilot — Local RAG Support Assistant with Citations and Human Escalation

Python, FastAPI, Ollama, Qdrant, Streamlit, Docker, pytest | Oct 2026 | github.com/VIKAS-PURBIA/support-copilot

- Built an agentic RAG pipeline (router, rewriter, hybrid retriever, grader, answerer, verifier) that answers support questions from a help-center knowledge base and past tickets with cited sources.
- Implemented hybrid search combining BM25 and dense vector retrieval with Reciprocal Rank Fusion, and role-based access control as a metadata filter inside vector queries.
- Designed four independent refusal gates (retrieval score, grader, citation check, verifier) with automatic human-handoff escalation notes and per-request agent traces with latency metrics.
- Hardened LLM prompts against prompt injection by treating retrieved text as data, and built a 50+ question evaluation harness measuring retrieval and answer quality, including unanswerable queries.

Multiclass Fish Image Classification — Deep Learning Web App

Python, TensorFlow, Keras, CNN, Transfer Learning, Streamlit | Sep 2025 | github.com/VIKAS-PURBIA/Multiclass-Fish-Image-Classification

- Trained a CNN from scratch and fine-tuned MobileNetV2 and EfficientNetB0 pretrained models on a dataset of 10,000+ fish images across multiple species.
- Achieved 92% classification accuracy through data augmentation and systematic hyperparameter tuning.
- Evaluated models with accuracy, precision, recall, F1-score and confusion matrix; deployed the best model as a Streamlit web app returning species predictions with confidence scores.

Zomato Restaurant Clustering and Sentiment Analysis

Python, scikit-learn, NLP | Jul 2025 | github.com/VIKAS-PURBIA/Zomato_Restaurant_Clustering_and_Sentiment_Analysis

- Engineered clustering algorithms to segment restaurants by cuisine and customer ratings, and performed NLP-based sentiment analysis on customer reviews to uncover experience and service insights.
- Visualized clustering and sentiment outcomes to guide strategic business decisions.

Customer Segmentation and Product Recommendations in E-Commerce (Shopper Spectrum)

Python, Machine Learning, RFM analysis | Jun 2025 | github.com/VIKAS-PURBIA/Shopper-Spectrum-Customer-Segmentation-and-Product-Recommendations-in-E-Commerce

- Built ML-driven RFM customer segmentation models and collaborative filtering product recommendation systems that improved e-commerce engagement and cross-selling.
- Measured project efficacy through a significant rise in average order value.

Local Food Wastage Management System

Python, SQL, Data Analysis | Sep 2025 | github.com/VIKAS-PURBIA/Local-Food-Wastage-Management-System

- Created a system to monitor and mitigate food waste by analyzing wastage trends, contributing to effective policy recommendations.
- Partnered with local organizations to improve food recovery efforts, achieving a 30% reduction in waste.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science

Bikaner Technical University — 2023 – 2027 | CGPA: 8.04